VirusPlatform

Data acquisition

Data resources

- Public data search (SRA + GSA)
- Bulk download (aria2c + sracha)
- Genome / host genome download (NCBI datasets)
- Open-Virome public viromes
- Sample virome prescreening
- Local database construction

Virus Detection & Identification

Sample processing

- Sample creation & batch import
- One-click preprocessing
- Read QC (fastp)
- Format conversion (fq2fa)
- Host read removal (kunpeng)

Quantification & consensus

- Known virus ID & quantification (salmon)
- Coverage depth & BAM
- Consensus sequence calling
- Variant analysis

Identification & assembly

- One-click pipeline (stage 2-4)
- Metaviral assembly (SPAdes)
- Candidate validation (BLASTx / BFVD / viroid)
- Assembly re-identification
- Host prediction (ICTV cascade)

Annotation & genomics

Genome annotation

- ORF prediction (pyrodigal)
- Functional annotation (DIAMOND / Pfam / BFVD)
- Homology search (BLASTN / BLASTX)
- Genome maps (gbdraw / DFV)

Comparative genomics

- Reference acquisition & CDS/PEP export
- Multiple alignment (MAFFT + trimAl)
- Tree inference (RAxML-NG / IQ-TREE / FastTree)
- SDT pairwise identity (exact engine)

Evolution & Phylodynamics

Phylodynamics

- Metadata governance & alignment QC
- Temporal signal & dating
- ML molecular clock (TreeTime)
- Skyline & Bayesian skyline plot
- RTT / DRT regression
- Phylogeography (mugration + migration GIF)
- Recombination (RDP5)
- BEAST interface (Markov jump / MCC)
- One-click dating - geography - tracing

Assay design

Assay & control design

- Primer design (primer3 + thermodynamics)
- CRISPR detection assay design
- dsRNA design pipeline (off-target scoring)
- miRNA target prediction (5 engines)

Results, Sharing & Platform

Results & reporting

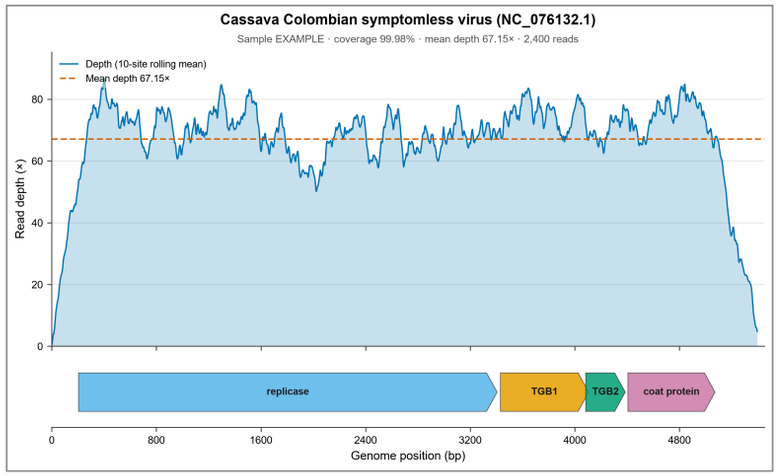
- Sample & module result center
- Tree / SDT / MSA viewers
- Interactive reports (plotly / pycirclize)

Tracing & submission

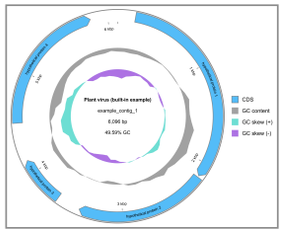
- LOGAN tracing (23.4M public SRA runs)
- NCBI submission (GenBank / BioSample / MIUVIG)

Platform capabilities

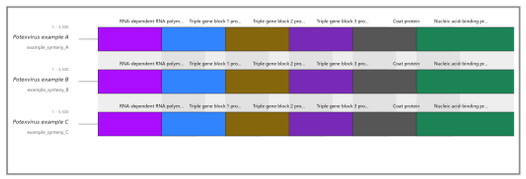
- AI assistant (interpret / broadcast / gate)
- Task center (self-learning ETA)
- Full CLI (40 subcommands)
- Bilingual interface (EN / CN)



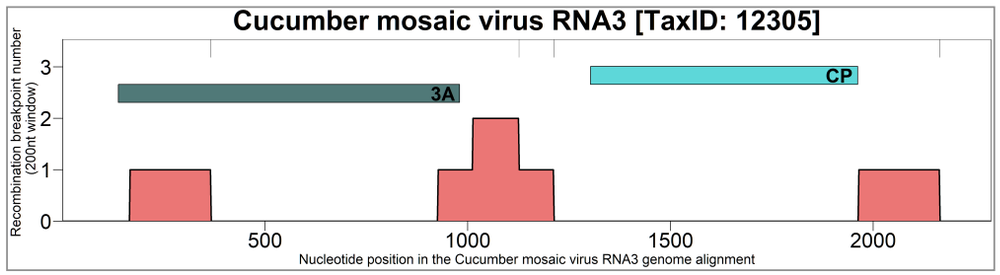
Depth plot (kvsuite)



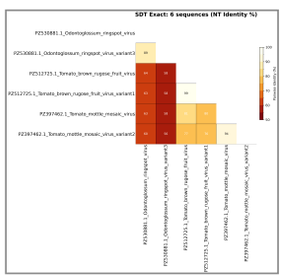
Genome map (gbdraw)



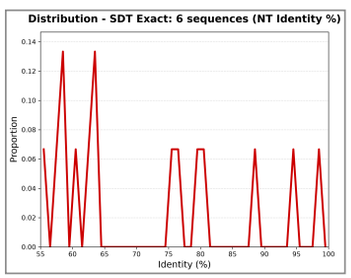
Synteny comparison



Recombination (RDP5)



SDT identity heatmap



Identity distribution